

Question 4

Q: Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Theoretical Concept: Pixel vs Intensity Resolution

Pixel Resolution (Spatial): Determined by sampling. It dictates the physical details visible (e.g., identifying a tiny tumor). Higher spatial resolution = more pixels.

Intensity Resolution (Radiometric): Determined by quantization. It dictates the number of gray levels (e.g., 8-bit = 256 shades). In medical imaging (like X-rays), high intensity resolution (12-bit or 16-bit) is crucial to differentiate subtle tissue densities that look identical in 8-bit images.

OpenCV Python Solution

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# Unique Python Program
# Pixel Resolution vs Intensity Resolution Demonstration
# Runs in Google Colab

import cv2
import numpy as np
import matplotlib.pyplot as plt

# Create a synthetic medical-like grayscale image
img = np.zeros((256, 256), dtype=np.uint8)

# Draw circles representing organs/tissues
cv2.circle(img, (128, 128), 80, 180, -1)
cv2.circle(img, (128, 128), 40, 230, -1)
cv2.rectangle(img, (30, 180), (90, 240), 120, -1)

# ----- Pixel Resolution -----
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low_pixel = cv2.resize(img, (64, 64), interpolation=cv2.INTER_AREA)
pixel_result = cv2.resize(low_pixel, (256, 256),
interpolation=cv2.INTER_NEAREST)

# ----- Intensity Resolution -----
levels = 16
intensity_result = (img // (256 // levels)) * (256 // levels)

# Display Results
titles = [
    "Original Image",
    "Reduced Pixel Resolution",
    "Reduced Intensity Resolution"
]

images = [img, pixel_result, intensity_result]

plt.figure(figsize=(12,4))

for i in range(3):
    plt.subplot(1,3,i+1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")

plt.show()

print("Observation:")
print("1. Lower pixel resolution decreases image sharpness.")
print("2. Lower intensity resolution decreases image contrast.")
print("3. In medical imaging, both high pixel and high intensity
resolution improve diagnosis.")

```

Expected Output

The output shows two gradients. The 8-bit gradient is perfectly smooth. The 3-bit gradient is broken into rigid vertical bars (false contours), demonstrating loss of intensity resolution.